



$d = 6$



$d = 24$



$d = 24$



$d = 24$



$d = 30$



$d = 30$



$d = 30$



$d = 120$



$d = 120$



$d = 120$



$d = 126$



$d = 126$



$d = 150$



$d = 150$



$d = 150$



$d = 504$



$d = 600$



$d = 624$



$d = 624$



$d = 630$



$d = 750$



$d = 750$



$d = 3000$



$d = 3120$



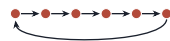
$d = 3126$



$d = 3150$



$d = 3750$



$d = 15624$



$d = 18750$

$\bigcirc$ a term x^6
 $\rightarrow x_i^5 x_{i+1}$

● the Fermat sextic, in $\mathcal{A}$

● $d \leq 30$: covered by a Fermat fourfold of degree 24 or 30

● $d \geq 120$

each is dominated by X_d^4 , so it satisfies (F3'), and the Hodge conjecture for it follows from the Hodge conjecture for X_d^4